

CV2102-1 – STRENGTH OF MATERIALS
(CV)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. A bar of diameter 30mm is subjected to a tensile load such that the measured extension on a gauge length of 200mm is 0.09mm and the change in diameter is 0.0045mm. Calculate the Poissons ratio?
A) 1/5
B) 1/3
C) 1/6
D) 1/4
2. The tensile test is carried on which material.
A) Malleable
B) Ductile
C) Plastic
D) Brittle
3. Hooke's law is valid up to
A) Elastic limit
B) Breaking point
C) Yield point
D) Proportional limit
4. Poisson's ratio is the ratio of
A) Longitudinal strain to lateral strain
B) Lateral strain to longitudinal strain
C) Lateral stress to longitudinal stress
D) Longitudinal stress to lateral stress
5. Principal stress acts on
A) Principal planes
B) Shear plane
C) On all planes
D) None of the above
6. If a part is constrained to move and heated, it will develop
A) Compressive stress
B) Principal stress
C) Tensile stress
D) Shear stress
7. Principal plane is also called as
A) Shear free plane
B) Plane of maximum shear
C) Plane of shear
D) Direct plane
8. Angle(Theta) made by a plane in a 2D stress system is equal to _____ in Mohar's circle
A) 0.5 theta
B) Theta
C) 2 Theta
C) 0.33 Theta
9. Simple bending equation is
A) $M/I = E/R = F/Y$
B) $M/I = R/E = F/Y$
C) $I/M = E/R = Y/F$
D) $M/I = R/E = Y/F$
10. Maximum shearing stress in a beam is at
A) Neural axis
B) Extreme fibre
C) Action of loading
D) Mid span
11. At the neutral axis, bending stress is
A) Zero
B) Maximum
C) Minimum
D) Constant

12. The layer at which neither compression nor tension acts is called
 A) Neutral layer B) Top layer
 C) Bottom layer D) Middle layer
13. In thin cylinders, the thickness should be _____ times of internal diameter
 A) 1/15 B) 1/10
 C) 1/20 D) 1/25
14. The hoop stress is also termed as
 A) Compressive B) Direct
 C) Circumferential D) Tensile
15. Radial stress in thin cylinders is
 A) Uniform B) Parabolic
 C) Neglected D) Linear
16. Which cylinder can also be subjected to external pressure
 A) Thick cylinder B) Thin cylinder
 C) Both thin and thick cylinder D) None of the above
17. What is the effective length of a column at both ends fixed?
 A) $2L$ B) $L/3$
 C) $L/2$ D) L
18. Long columns fail due to _____
 A) Direct stress B) Buckling stress
 C) Lateral stress D) Tensile stress
19. What is the effective length of a column at both ends hinged?
 A) $2L$ B) $L/3$
 C) $L/2$ D) L
20. What is the effective length of a column with one end fixed and other end free?
 A) $2L$ B) $L/3$
 C) $L/2$ D) L

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Define young's modulus and bulk modulus and write an expression for relation between them.
 b) A bar of sides $0.040\text{m} \times 0.040\text{m}$ is subjected to an axial pull of 160000N . Decrease in width is 0.005mm . The extension over a gauge length of 200mm is 0.1mm . Calculate the elastic constants E, G, K and Poisson's ratio.
 c) The state of stress in a two dimensionally stressed body is as shown in Figure. Q 1 (c). Determine principal stresses and maximum shear stress.

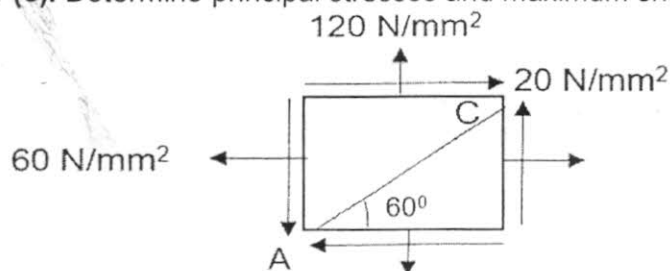


Figure. Q 1 (c)

2. a) Define the following with mathematical expressions i) Poisson's ratio ii) Thermal stress iii) Rigidity modulus
 b) Graphically represent a typical stress-strain relation for mild steel and mark all the salient points on the curve

Marks	BT*	CO*	PO*
3	L*2	1	1,2
5	L3	1	2
8	L3	2	1,2
3	L2	1	1,2
5	L2	1	1,2

- c) The state of stress in a 2D system is as shown in **Figure. Q 2 (c)**. Determine principal stresses and maximum shear stress.

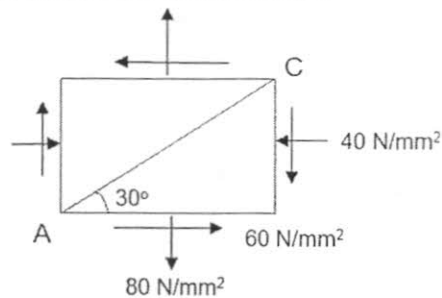


Figure. Q 2 (c)

3. a) A bronze bar is fastened between a steel bar and an aluminum bar as shown in **Fig. Q 3 (a)**. Axial loads are applied at the positions indicated. Find the largest value of P that will not exceed an overall deformation of 3.0 mm, or the following stresses: 140 MPa in the steel, 120 MPa in the bronze, and 80 MPa in the aluminum. Assume that the assembly is suitably braced to prevent buckling. Use $E_{st} = 200$ GPa, $E_{al} = 70$ GPa, and $E_{br} = 83$ GPa.

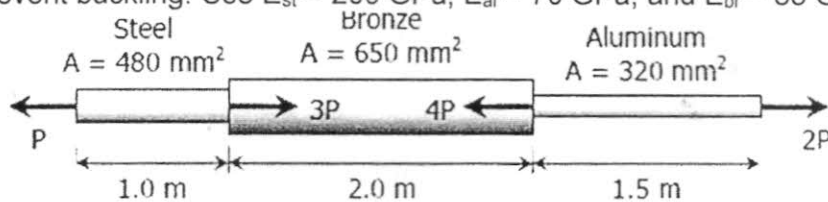


Figure. Q 3 (a)

- b) The state of stress in a 2D system is as shown in **Figure. Q 1 (c)**. and **Figure. Q 2 (c)**. Determine the normal and tangential stresses on plane AC in **Figure. Q 1 (c)**. and **Figure. Q 2 (c)**.

Unit – II

4. a) Two wooden planks 150mm X 50mm each are connected to form a T section of beam. If a moment of 3.4 kN-m is applied about the horizontal neutral axis with tension below neutral axis. Find the stresses at extreme fibers of the section. Draw the stress distribution on the cross-section.
- b) A thick cylindrical pipe with outer diameter 300mm and inner diameter 200mm is subjected to an internal fluid pressure of 14N/mm². Determine the maximum hoop stress developed and radial pressure and sketch the variation of stresses over the thickness.
5. a) Write any four assumptions made in simple bending theory and explain the terms i) Flexural rigidity ii) Section modulus.
- b) A thin cylindrical shell 1m in diameter and 3m long has a 10mm thickness subjected to an internal pressure of 3Mpa. Determine the changes in volume and maximum shear stress in the shell. Assume $E_s = 210$ Gpa and $\mu = 0.3$.
6. a) Obtain the expression for maximum deflection of a simply supported beam of length "L" and subjected to a UDL of intensity "w/m" throughout the beam.
- b) Write the lame's equation for thick cylinder and mention two differences between thin and thick cylinders.

Unit – III

7. a) A 2m long column has a square cross-section of sides 40mm. Taking factor of safety 4, determine the safe load for the end conditions i) Both ends are hinged ii) One end is fixed and other end is free iii) Both ends are fixed iv) One end is fixed and other end is hinged. Take $E = 210$ Gpa.

- b) A solid round bar of 60mm diameter and 2.5m long is used as a strut. Find the safe compressive load for the strut for the following conditions: i) both ends are hinged ii) both ends are fixed. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and factor of safety is 3.
8. a) Define effective length of a column and mention the values for various end conditions.
- b) Write the expression for Rankine's formula and Euler's formula. Distinguish long column and short column.

8	L3	5	1,2
8	L3	5	1,2
8	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
